TakeSure

Confidence before taking something

An Ayurveda × Allopathy herb–drug interaction checker for Indian patients, doctors and the Ministry of Ayush

HACK@BVP 8.0 · Abstract Round

Team Bug Hunters

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Domain: Health & Well-Being (AI4Impact secondary) · SDG 3 · SDG 9 · SDG 10

01 THE PROBLEM

- **Two medical systems, zero cross-checking.** A thyroid patient on Thyronorm adds ashwagandha; a transplant patient on tacrolimus drinks giloy kadha; a cardiac patient on warfarin takes guggul.
- **Each pair can cause harm:** ashwagandha can raise thyroid hormone, giloy can oppose immunosuppression, guggul can increase bleeding.
- **Nobody asks, nobody tells.** The doctor rarely asks about herbs, the patient rarely mentions them, the chemist has no tool, the label says nothing.
- **Existing tools do not cover it.** Medscape, Drugs.com and 1mg check only allopathic × allopathic pairs; Western herb databases are English-only, paid and do not know Indian brands like Himalaya Ashvagandha or Dabur Giloy Ghanvati.

37% of hospital patients use allopathic and herbal medicines together

65% of herbal users do not know the dose, side effects or interactions

19% of Indian patients ever disclose herbal use to their doctor

1,473 ASU&H adverse-event reports received nationally since 2018

02 PROPOSED SOLUTION

- **Scan both.** The patient photographs the allopathic prescription and the Ayurvedic product label (or types the names); every name is confirmed before checking.
- **Match Indian names.** Brand → generic (Thyronorm → levothyroxine) and product → herbs (Chyawanprash → amla, ashwagandha…) with Sanskrit / Hindi / Latin synonyms.
- **Decide from data.** A rule engine checks every herb–drug pair against a curated, cited knowledge base: GREEN / AMBER / RED / NOT IN DATABASE, with timing advice.
- **Explain simply.** AI rewrites only the matched evidence into plain Hindi or English, with sources and three questions to ask the doctor.
- **Report in one tap.** A one-page PDF for the doctor and a pre-filled suspected-ADR report in the Ministry of Ayush format for the Ayush Suraksha portal.

03 APPROACH

Inputs
Prescription photo + Ayurvedic product photo (or typed names)

▼

1 · OCR
read both images; every name confirmed by the user

▼

2 · Match
brand → generic; product → herbs; Sanskrit/Hindi/Latin synonyms

▼

3 · Knowledge base
cited herb–drug rows with severity + evidence grade

▼

Rule engine
decides GREEN / AMBER / RED / NOT IN DB — AI does not

▼

4 · AI explains
plain Hindi/English, sources, 3 questions for the doctor

▼

Outputs
Doctor PDF · ADR report (Ministry format) · analytics

Worst row wins; an unknown pair is never shown as green; grade-C evidence caps at AMBER; the answer is identical with the AI switched off.

"The verdict comes from published data; AI only explains it."

Feature	Medscape · Drugs.com · 1mg	MSKCC · Natural Medicines	TakeSure
Ayurveda × allopathy pair check	✗	~ single herbs	✓ every pair
Indian brands & Hindi	~	✗	✓ brands + synonyms
Source on every verdict	✓	✓	✓ + evidence grade
ADR report to Ministry of Ayush	✗	✗	✓ one tap

Our assessment from each tool's public documentation.

Stack. Next.js + TypeScript PWA (Hindi/English UI) · FastAPI + Pydantic REST API · PostgreSQL in production, SQLite offline · Tesseract OCR · RapidFuzz matching (≥85 accept, 70–85 ask the user) · Groq LLM, JSON-schema output, temperature 0 · ReportLab PDFs. No model training; runs fully offline on a laptop.

Knowledge base. 40–60 high-frequency herb × drug pairs (15 common herbs × 12 drug classes) curated from PubMed, NIH LiverTox, MSKCC About Herbs, WHO monographs and the Ayurvedic Pharmacopoeia of India, graded on the Stockley's model. Decision support only: "ask your doctor", never diagnosis or dosing.

04 EXPECTED IMPACT

Patients & caregivers A sourced answer before the first dose, in their own language, instead of silence or a guess. Free for citizens.

Doctors, vaidyas & chemists Finally learn what patients actually take; a QR on the chemist's bill or an API inside hospital and telemedicine systems makes the check routine.

Ministry of Ayush Structured, reproducible ADR reports and real co-use data from the catchment of its 99 peripheral pharmacovigilance centres.

Novelty Nothing public today checks an Ayurvedic product against an allopathic prescription. TakeSure builds the missing bridge table: Indian brands × Ayurvedic herb synonyms × graded evidence, open and auditable. Honest by design: grades, timing gaps instead of alarms, "not in database" instead of guesses.